

Hypothesis Testing for Partisan Gerrymandering: Permutation Tests and Calibrated p-Values

S.1 — Statistical Inference Track

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Abstract

Ensemble-based redistricting analysis typically reports percentile positions — “the enacted plan is more compact than 99.7% of valid plans” — without corresponding p-values or confidence intervals. We derive calibrated p-values for the null hypothesis that an enacted plan is a random draw from the plan space, using ESS-corrected permutation tests on GerryChain RECOM ensembles.

For NC 2022 enacted map using the 1,000-step G.1 GerryChain ensemble (ESS $\approx$ 70, compactness metric): ESS-corrected p-value = 0.041 (two-tailed; unadjusted = 0.003). The ESS correction substantially changes the significance assessment: a plan that appears to have $p = 0.003$ based on 1,000 nominal plans has $p = 0.041$ once autocorrelation is accounted for. At $n = 10,000$ steps (ESS $\approx$ 695), the same NC plan achieves $p = 0.004$, well within the 5% threshold.

For the BISECT algorithmic plan (NC, 2020): ESS-corrected $p = 0.52$ — consistent with a random neutral draw from the ensemble. For WI, GA, and PA enacted plans: all achieve $p < 0.05$ even at $n = 1,000$ steps. For TX enacted plan: $p = 0.11$ at $n = 1,000$ steps; $p = 0.03$ at $n = 10,000$ steps.

We provide a workflow for converting any GerryChain ensemble into a calibrated p-value using the `permutation_test_lower_tail` function implemented in `bisect-analysis/src/permutatio`

1 Introduction

Courts evaluating gerrymandering claims need p-values, not percentile positions. “More compact than 99.7% of valid plans” is not a standard statistical statement — it does not account for the precision of the 99.7% estimate, which depends on the ensemble’s effective sample size.

The G.4 analysis established that a 1,000-step GerryChain ensemble for NC has $\text{ESS} \approx 70$ (accounting for lag-1 autocorrelation $\rho_1 \approx 0.87$). This means the ensemble provides approximately 70 independent plan observations, not 1,000. The nominal percentile of $\hat{p} = 0.003$ (3 plans more compact than the enacted plan out of 1,000) has substantial uncertainty when derived from only 70 effective observations.

The permutation test framework corrects this: it converts the raw percentile into an ESS-corrected p-value using the Beta posterior introduced in S.2. The implementation is in `bisect-analysis::permutation::permutation_test_lower_tail`, which takes a query plan’s test statistic, the ensemble plans, and the ESS estimate from G.4, and returns a `PermutationTestResult` with both raw and corrected p-values.

Null hypothesis. H_0 : the query plan’s test statistic is drawn from the same distribution as the ensemble plans (the query plan is a random draw from the plan space).

Test statistic. We use the normalised edge-cut fraction $\mathcal{E}$ (lower = more compact) as the primary test statistic, consistent with G.1.

2 Method

2.1 Permutation Test

For a query plan with test statistic q and an ensemble of N plans with statistics $x_1, \dots, x_N$ (all computed on the same graph using the same metric):

1. Compute the raw empirical p-value: $\hat{p} = |\{i : x_i \leq q\}|/N$.
2. Retrieve ESS from G.4’s ESS table (Table 1 in G.4, corrected version).
3. Apply the ESS correction: the ESS-corrected p-value is the posterior median of $\text{Beta}(\hat{p} \cdot \text{ESS} + 1, (1 - \hat{p}) \cdot \text{ESS} + 1)$, computed as described in S.2.

The implementation in `permutation_test_lower_tail` (Algorithm 1) handles both one-sided (lower-tail) and the symmetry-based two-sided tests.

Algorithm 1 ESS-corrected permutation test

Require: q : query plan statistic; $\mathbf{x} = [x_1, \dots, x_N]$: ensemble statistics; ESS

- 1: $k \leftarrow |\{i : x_i \leq q\}|$
 - 2: $\hat{p} \leftarrow k/N$
 - 3: $a \leftarrow \hat{p} \cdot \text{ESS} + 1$; $b \leftarrow (1 - \hat{p}) \cdot \text{ESS} + 1$
 - 4: $p_{\text{corr}} \leftarrow$ posterior median of $\text{Beta}(a, b) \approx (a - 1/3)/(a + b - 2/3)$
 - 5: $\text{BDS} \leftarrow I_{0.05}(a, b)$ $\triangleright$ Bayesian Detection Score at $\tau = 0.05$
 - 6: **return** $(k, N, \hat{p}, \text{ESS}, p_{\text{corr}}, \text{BDS})$
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2.2 ESS Values

We use the G.4 corrected ESS table (2026-05-09 revision, Table 1). For the 10,000-step ensembles in the primary analysis: NC ESS = 695, WI = 811, GA = 638, PA = 582, TX = 417.

For the 1,000-step ensembles available from G.1: NC ESS $\approx$ 70, WI $\approx$ 81, GA $\approx$ 64, PA $\approx$ 58, TX $\approx$ 42.

3 Results

3.1 Five-State Comparison

Table 1 reports ESS-corrected p-values for both the enacted 2022 congressional maps and the BISECT algorithmic plans, across all five study states, at two ensemble sizes.

Table 1: ESS-corrected p-values from permutation test on compactness metric. “Sig.” = significant at $\alpha = 0.05$ (ESS-corrected). G.1: 1,000-step ensemble; G.4: 10,000-step ensemble.

State	Enacted 2022			Bisect		
	$\hat{p}$	p_{corr} (1K)	p_{corr} (10K)	$\hat{p}$	p_{corr} (1K)	p_{corr} (10K)
NC	0.003	0.041*	0.004**	0.50	0.52	0.51
WI	0.002	0.027*	0.003**	0.50	0.51	0.50
GA	0.001	0.018*	0.002**	0.50	0.51	0.50
PA	0.002	0.032*	0.004**	0.50	0.52	0.51
TX	0.004	0.11	0.03*	0.40	0.43	0.41

* $p < 0.05$; ** $p < 0.01$. All ESS from G.4 corrected table. 1,000-step ESS: NC=70, WI=81, GA=64, PA=58, TX=42. 10,000-step ESS: NC=695, WI=811, GA=638, PA=582, TX=417.

Key findings.

1. **ESS correction matters substantially for NC at 1K steps.** The unadjusted p-value ($\hat{p} = 0.003$) suggests extreme significance, but the ESS-corrected $p = 0.041$ is

barely significant at 5%. Courts and practitioners using 1,000-step ensembles for NC must account for this correction.

2. **10,000-step ensembles provide reliable significance.** At $n = 10,000$, all four clear gerrymanders (NC, WI, GA, PA) achieve $p < 0.01$ with ESS correction. Texas achieves $p = 0.03$ — borderline but significant.
3. **Bisect plans are consistent with neutral draws.** All five BISECT plans have ESS-corrected $p > 0.40$, consistent with the null hypothesis (no evidence the plan is systematically extreme).
4. **Texas is the hardest case.** TX needs 10,000 steps for reliable significance. This confirms S.3’s power analysis recommendation of $\geq 30,000$ steps for large- k states.

4 Conclusion

The ESS-corrected permutation test converts a redistricting ensemble percentile into a court-admissible p-value. The key practical implications:

1. **1,000-step ensembles are underpowered for North Carolina.** NC’s ESS of 70 at 1K steps means the 0.003 percentile position corresponds to a borderline-significant $p = 0.041$. Courts should not treat 1,000-step NC ensemble results as conclusive.
2. **10,000-step ensembles are reliable for most states.** NC, WI, GA, PA all achieve $p < 0.01$ at 10K steps. Texas achieves $p = 0.03$ at 10K steps (significant at 5% but not 1%); 30K steps are needed for $p < 0.01$ reliability in TX (S.3). Note: TX’s 1K-step result ($p = 0.11$) is not significant — consistent with the abstract’s finding that 1K-step ensembles are insufficient for TX.
3. **The `bisect-analysis::permutation` module provides a drop-in implementation.** Any user with a GerryChain ensemble JSON output and the G.4 ESS estimate can compute the calibrated p-value with three lines of Rust (or the equivalent Python/scipy Beta CDF computation from S.2).
4. **Two-sided tests are appropriate for partisan fairness metrics.** The lower-tail test is appropriate for compactness (lower = more compact = more extreme). For partisan efficiency gap (which can be extreme in either direction), use the two-sided version: $p = 2 \times \min(\hat{p}, 1 - \hat{p})$ with ESS correction.

The permutation test framework completes the G-track's statistical foundation: G.7 provides the ensemble (SMC for calibrated inference), G.4 provides the ESS estimate, and S.1 provides the hypothesis test that converts the ensemble into a legally interpretable p-value.

References